Lab 1A: String Induction

The following problems ask you to prove some "obvious" claims about recursively-defined string functions. In each case, we want a self-contained, step-by-step induction proof that builds on formal defintions and prior results, *not* on intuition. In particular, your proofs must refer to the formal recursive defintions of string length and string concatenation:

$$|w| := \begin{cases} 0 & \text{if } w = \varepsilon \\ 1 + |x| & \text{if } w = ax \text{ for some symbol } a \text{ and some string } x \end{cases}$$

$$w \bullet z := \begin{cases} z & \text{if } w = \varepsilon \\ a \cdot (x \bullet z) & \text{if } w = ax \text{ for some symbol } a \text{ and some string } x \end{cases}$$

You may freely use the following results:

Lemma 1. *$w \bullet \varepsilon = w$ for all strings w.*

Lemma 2. *$|w \bullet z| = |w| + |z|$ for all strings w and z.*

Lemma 3. *$(w \bullet y) \bullet z = w \bullet (y \bullet z)$ for all strings w, y, and z.*

The **reversal w^R** of a string w is defined recursively as follows

$$w^R := \begin{cases} \varepsilon & \text{if } w = \varepsilon \\ x^R \bullet a & \text{if } w = ax \text{ for some symbol } a \text{ and some string } x \end{cases}$$

For example, $\texttt{STRESSED}^R = \texttt{DESSERTS}$ and $\texttt{WTF374}^R = \texttt{473FTW}$.

1. Prove that $|w| = |w^R|$ for every string w.

   Assume that $|x| = |x^R|$ for every string x shorter than w.

   Case 1: Suppose that $w = \varepsilon$.

   Then,

   $$\begin{aligned} |w^R| &= |\varepsilon^R| \\ &= |\varepsilon| \quad \text{by definition of reversal} \\ &= |w| \end{aligned}$$

   Case 2: $w = ax$ for some symbol a and string x.

   Then,

$$
\begin{aligned}
|w^R| &= |x^R \bullet a| && \text{by definition of reversal} \\
&= |x^R| + |a| && \text{by definition of } \textbf{Lemma 2} \\
&= |x^R| + 1 && \text{by definition of length (since } |a| = |a\varepsilon| = 1 + |\varepsilon| = 1) \\
&= |x| + 1 && \text{by defintion of inductive hypothesis} \\
&= |ax| && \text{by defintion of length} \\
&= |w|
\end{aligned}
$$

2. Prove that $(w \bullet z)^R = z^R \bullet w^R$ for all strings w and z.

   Assume that $(x \bullet z)^R = z^R \bullet x^R$ for every string x shorter than w (and every string z)

   Case 1: Suppose that $w = \varepsilon$.

   Then,

$$
\begin{aligned}
(w \bullet z)^R &= (\varepsilon \bullet z)^R \\
&= z^R && \text{by definition of concatenation} \\
&= z^R \bullet \varepsilon && \text{by definition of } \textbf{Lemma 1} \\
&= z^R \bullet \varepsilon^R && \text{by definition of reversal} \\
&= z^R \bullet w^R
\end{aligned}
$$

   Case 2: Suppose that $w = ax$ for some symbol a and string x.

   Then,

$$
\begin{aligned}
(w \bullet z)^R &= (ax \bullet z)^R \\
&= (a \cdot (x \bullet z))^R && \text{by definition of concatenation} \\
&= (x \bullet z)^R \bullet a && \text{by definition of reversal} \\
&= \left(z^R \bullet x^R\right) \bullet a && \text{by definition of inductive hypothesis} \\
&= z^R \bullet \left(x^R \bullet a\right) && \text{by definition of } \textbf{Lemma 3} \\
&= z^R \bullet w^R && \text{by definition of reversal}
\end{aligned}
$$

3. Prove that $\left(w^R\right)^R = w$ for every string w

   Assume that $\left(x^R\right)^R = x$ for every string x shorter than w.

   Case 1: Suppose that $w = \varepsilon$.

   Then,

$$
\begin{aligned}
\left(w^R\right)^R &= \left(\varepsilon^R\right)^R \\
&= \varepsilon^R && \text{by definition of reversal} \\
&= \varepsilon && \text{by definition of reversal} \\
&= w
\end{aligned}
$$

Case 2: Suppose that $w = ax$ for some symbol a and string x

Then,

$$
\begin{aligned}
\left(w^R\right)^R &= \left((ax)^R\right)^R \\
&= \left(x^R \bullet a\right)^R && \text{by definition of reversal} \\
&= a^R \bullet \left(x^R\right)^R && \text{by Problem 2} \\
&= a^R \bullet x && \text{by the inductive hypothesis} \\
&= a \bullet x && \text{by definition of reversal } (a = a\varepsilon; a^R = \varepsilon^R \bullet a = \varepsilon \bullet a = a) \\
&= ax && \text{by definition of concatenation} \\
&= w
\end{aligned}
$$